

Quiz Master: Project Report

1. Author

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I'm passionate about building interactive and secure applications, always eager to learn and improve my skills

2. Description

This project involves developing a **Flask-based quiz game** with features like **score tracking, answer validation, and database management** using SQLite. The goal is to ensure a smooth user experience, integrate a proper backend, and enhance game interactivity

3. Technologies Used

- **Frontend:** HTML, CSS, JavaScript, Bootstrap, Jinja
- **Backend:** Flask, SQLite, SQLAlchemy
- **Security:** Flask-Login , Flask-Bcrypt
- **Graphs:** Matplotlib

I used **Flask with SQLite and SQLAlchemy** for efficient backend management, while **HTML, CSS, JS, Bootstrap, and Jinja** ensure a dynamic and responsive frontend. For security, **Flask-Login and Flask-Bcrypt** handle authentication and password protection.

4. Database Schem ER Diagram Link

https://drive.google.com/file/d/1J3QMq81coySRGJT2Ta6HG7_LerZSH9xJ/view?usp=sharing

This way designing of database optimal for **scalability and normalization**, ensuring structured relationships across all tables:

5. Graphs

I use Python's Matplotlib library for making graphs.

6. API Design

I created APIs for **user authentication, quiz management, and score tracking**, using Flask and SQLAlchemy to handle database interactions. I have also create resource API to provide data about subjects, chapters, quizzes, users etc,.

7. Architecture and Features

The project follows an **M-V-C structure**, with **controllers** in the controllers folder, **models** in models folder, and **templates** in templates folder for dynamic rendering. Static files like CSS and JS and images are stored in the static folder for frontend styling and interactivity.

The project includes **user authentication, quiz management, answer validation, and score tracking**, implemented using Flask and SQLAlchemy. Additional features like **performance graphs, leaderboards, and analytics** enhance user experience with dynamic data visualization.

8. Video

<https://drive.google.com/file/d/1LYOjkpuvu0rsy4iKGEEmHK4mwUEpJ-oQz/view?usp=sharing>